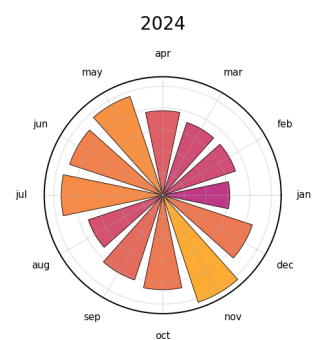
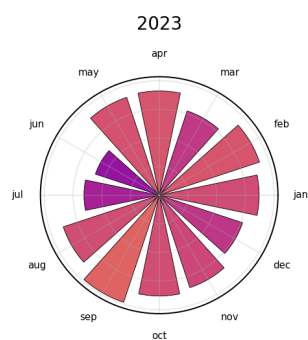
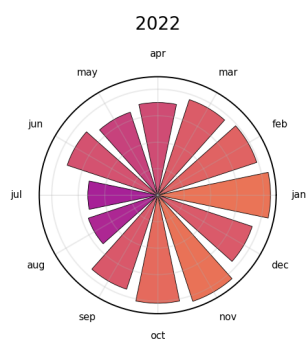
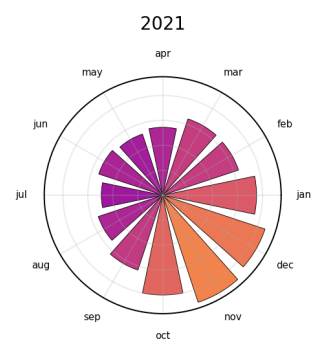
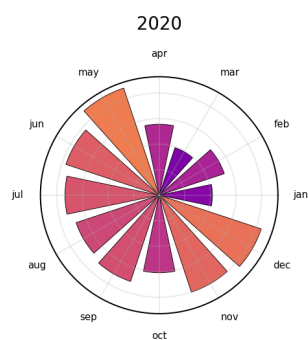
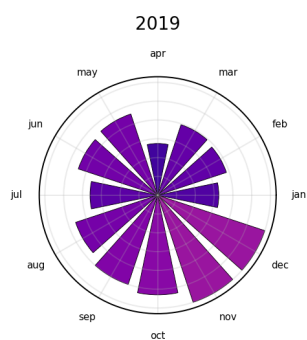
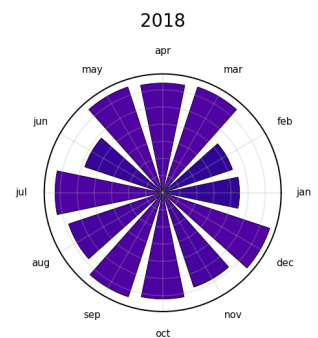
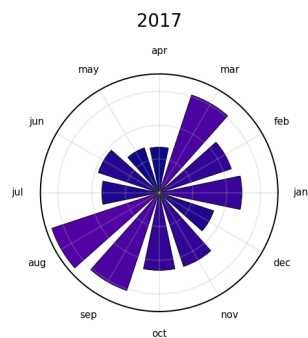
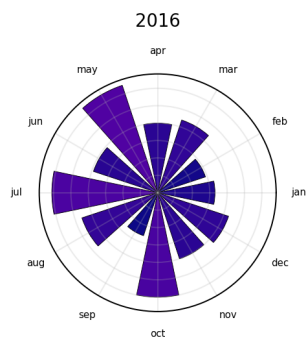


EDA cyber crime Leuven: reported incidents 2016-2025



2025 is featured in the updated analysis inside

Predicting Cybersecurity Incidents: Data Analysis



Before conducting the actual data analysis, this project starts with background research on cybercrime and the annual rhythm of major high-tech releases.

Research question: Do reported cybercrime incidents display recurring monthly or seasonal patterns between 2016 and 2025, and do any of these patterns coincide with periods of major technology releases or increased technology-related consumer activity?

January:

High-tech season begins with CES and major innovation announcements.

June:

Major software, hardware and developer announcements take place.

September to October:

Peak consumer-tech season with major device and platform releases.

November to December:

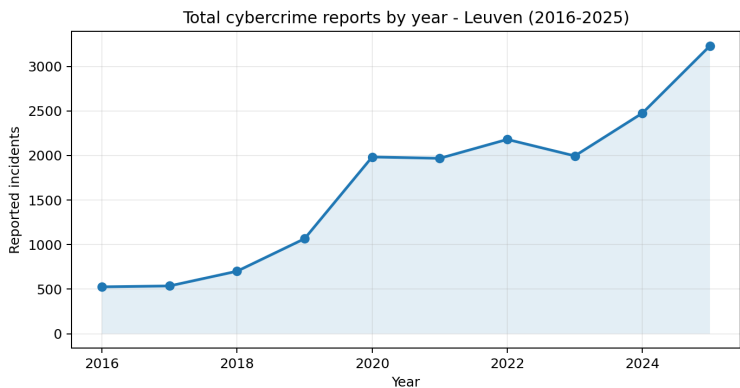
Holiday shopping, promotions and continued releases increase online activity.

The release-timing comparison is exploratory; timing alone does not establish causation.

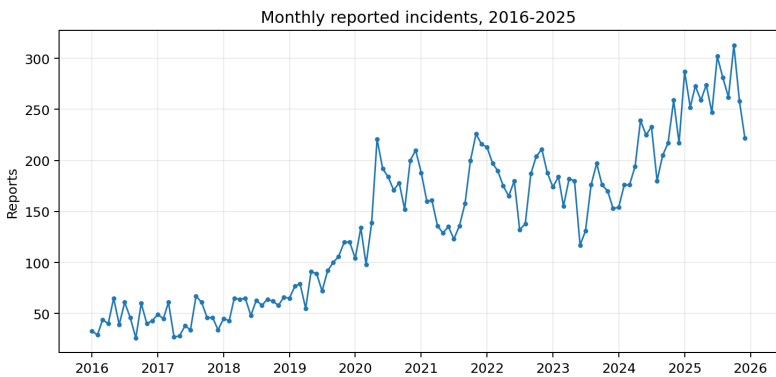
Key Insights

- A clear long-term increase is visible across the 2016-2025 series.
- Annual reports rose from 526 in 2016 to 3,230 in 2025.
- The increase is not perfectly linear: some years fall below the preceding year.
- Monthly totals vary across the calendar, with year-end months among the higher accumulated periods.

Details



Next Steps

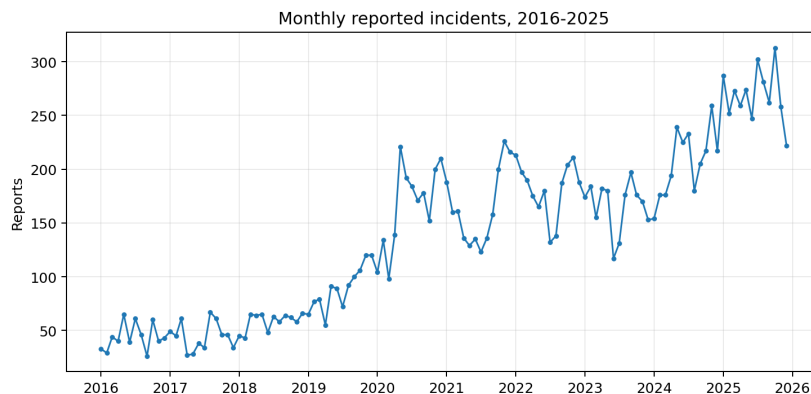


ISSUE / PROBLEM • RESPONSE • IMPACT

Monthly and seasonal patterns

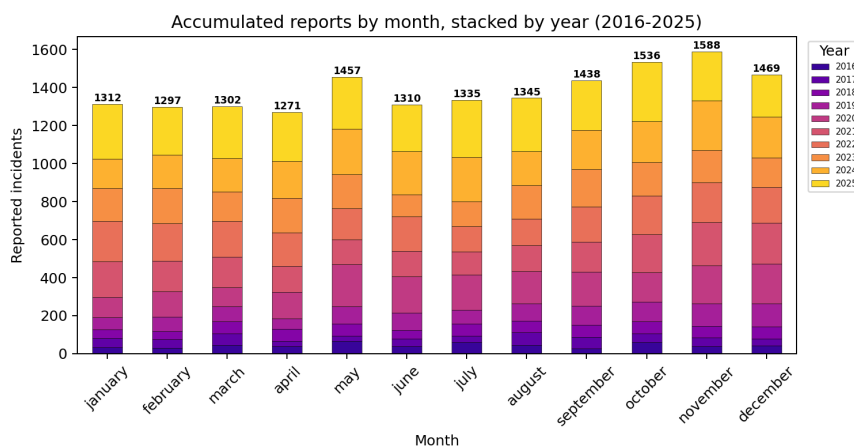
ISSUE / PROBLEM

Some months accumulate more reported incidents than others. The task is to visualize the pattern without confusing seasonality with the overall rise in cybercrime over time.



RESPONSE

Aggregate each calendar month across 2016-2025 and stack the bar by year. This preserves the total per month while showing how much each year contributes.



IMPACT

The chart makes both effects visible at once: accumulated monthly differences and the much larger contribution of recent years.

KEY INSIGHTS

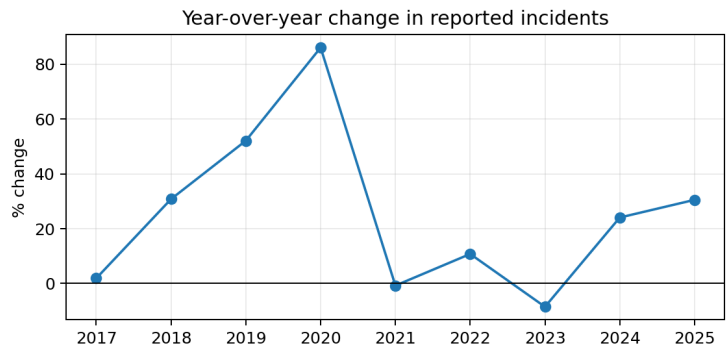
November has the highest accumulated total (1,588), followed by October (1,536) and December (1,469). April is the lowest accumulated month (1,271).

ISSUE / PROBLEM • RESPONSE • IMPACT

Visualize the yearly trend and cumulative scale

ISSUE / PROBLEM

Visualize how reported incidents changed over time and quantify the scale of the increase.



RESPONSE

Calculate annual totals, year-over-year change and the cumulative total across the 10-year dataset.

Year	Total	YoY change	Cumulative
2016	526		526
2017	536	+1.9%	1,062
2018	701	+30.8%	1,763
2019	1,066	+52.1%	2,829
2020	1,983	+86.0%	4,812
2021	1,968	-0.8%	6,780
2022	2,180	+10.8%	8,960
2023	1,995	-8.5%	10,955
2024	2,475	+24.1%	13,430
2025	3,230	+30.5%	16,660

IMPACT

The updated dataset contains 16,660 reported incidents. 2025 is the highest year in the current series.

KEY INSIGHTS

Reports increased from 526 in 2016 to 3,230 in 2025. The later years dominate the cumulative total, while year-over-year movement still includes temporary declines.

EDA Cybercrime Leuven

Data analysis of cybercrime reports 2016 - 2025

ISSUE / PROBLEM

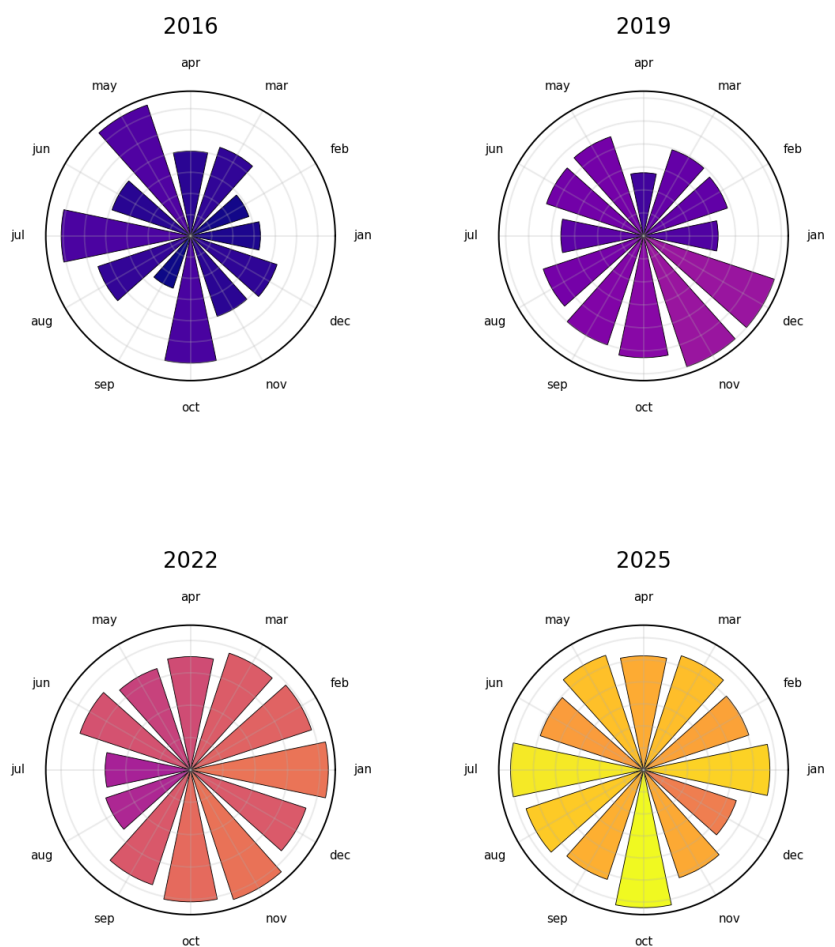
Reported cybercrime has increased substantially in the Leuven dataset.

RESPONSE

Use exploratory data analysis to compare monthly profiles across selected years and identify recurring or changing patterns.

IMPACT

The visual comparison supports awareness and further investigation, but does not establish why the patterns occur.



KEY INSIGHTS

Yearly trend: 2025 is the highest year in the updated dataset. Monthly pattern: November is the highest accumulated month across 2016-2025. Product-release timing remains a hypothesis for comparison, not a demonstrated causal explanation.

EDA Cybercrime Leuven

Project Overview

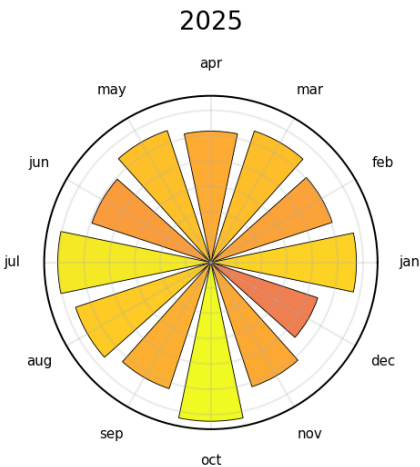
Project Overview

Conduct a simple exploratory data analysis to visualize trends and patterns in reported cybercrime data for Leuven, 2016-2025.

Details

Key Insights

- Yearly Trends - strong long-term increase, with 2025 highest.
- Monthly Patterns - November has the highest accumulated total.
- Seasonality - differences exist, but recent years also contribute much larger volumes.
- Product Releases - timing can be compared, but the present data does not prove causation.



Next Steps

Next, repeat the same EDA for other Belgian cities or police zones to test whether the Leuven pattern is local or more widespread. Add external variables such as product-release dates, vulnerability disclosures and phishing statistics for stronger hypothesis testing.

EDA cyber crime Leuven: reported incidents 2016-2025

Comparing with High-Tech Product Release Times

- **January**

CES and major innovation announcements create high public attention, but the Leuven data does not show the annual maximum here.

- **March-April**

Spring technology events can be compared with the data, but April is actually the lowest accumulated month in this dataset.

- **June**

Developer and software announcements provide a useful comparison point; the observed incident total is moderate relative to year-end.

- **September-October**

The accumulated totals rise into the autumn, with October among the highest months.

- **November-December**

November is the highest accumulated month, and December is also high. Holiday shopping and technology activity are plausible contextual factors, not proven causes.

Updated Findings

- Yearly Trends - reported incidents rise strongly over the 2016-2025 period, with 2025 highest.
- Monthly Patterns - November has the highest accumulated total; October and December are also high.
- Correlation with Product Releases - the current EDA can identify coinciding timing, but cannot establish a causal correlation from incident counts alone.
- Seasonal Variations - monthly differences should be interpreted alongside the much larger overall volume in recent years.

Recommendations

- Continue public-awareness campaigns during periods of high online and shopping activity.
- Maintain continuous monitoring rather than limiting attention to product-release windows.
- Compare Leuven with other Belgian cities using the same methodology.
- Add product-release dates, CVE disclosures, phishing volumes and other explanatory variables.
- Use statistical time-series methods before making predictive or causal claims.